

DIGITAL TRANSFORMATION & ARTIFICIAL INTELLIGENCE

Capstone Board Presentation — Project Report

CRED

Designing the AI-Native Enterprise

From a bill-pay app to India’s household financial intelligence layer

FIELD	DETAIL
Industry	Fintech / BFSI — India
Use case #1 (live)	AI Household Card Router
Live application	dtai-fintech-cred.vercel.app
GitHub repository	github.com/prathameshkumbhar1206-alt/DTAI-FINTECH-CRED
Team	DTAI Fintech · PGP Capstone (edit with your team’s names)
Report date	21 August 2026

A strategy without a working prototype is a proposal. This is not a proposal.

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1. Executive Summary

CRED should extend from a single-user bill-payment experience into household-level financial intelligence, starting with an AI-powered card router that ends manual reward comparison for multi-card families. Unlike a conventional strategy recommendation, this one is not speculative: the first use case is built, deployed to a public URL, and answering real purchase decisions today.

The underlying problem is structural. Indian financial decisions are made by households — spouses, parents, and dependents jointly manage income, cards, cover, and obligations — yet no major fintech product in the market treats the household as the unit of decision-making. CRED, despite owning the trust relationship with India's most creditworthy cardholders, currently sees only the individual member, not the family unit that member belongs to. This is both the company's clearest blind spot and its most defensible expansion opportunity, because the same card-linkage data CRED already collects for one person can be extended to a household with no new regulatory data class required.

The prototype built for this capstone, branded internally as the AI Household Card Router, demonstrates the opportunity concretely. Given a purchase described in plain English, it determines which credit card in a household — across every linked family member — earns the most reward value, and shows precisely which clause of that card's terms and conditions drove the answer. In live testing, the same ₹80,000 flight purchase produced a ₹9,200 swing in reward value depending on a single detail: whether the booking was made through the issuing bank's rewards portal or directly with the airline. That distinction is invisible to a person comparing headline cashback percentages, and it is exactly the kind of decision a household-level AI layer is positioned to catch.

Why the Board should approve this first

- It is the highest value × highest feasibility use case on the AI Opportunity Matrix (Section 5a) — every other candidate use case scores lower on at least one axis.
- It ships on data CRED already collects through its existing card-linkage flow. No new consent architecture or regulated data class is required to launch.
- Its pricing logic is fully deterministic. The AI model explains a decision that was already made by code, which keeps the associated AI risk contained to language generation rather than financial computation — the lowest-risk tier available on the Risk Assessment Matrix (Section 15a).
- For the demonstration household modelled in the prototype, optimal routing recovers an estimated ₹33,036 per year against the household's current default-card habit — a concrete, defensible, and independently recomputable number (Section 14).

The remainder of this report sets out the full strategic case — industry context, business challenges, the AI Opportunity Matrix, enterprise strategy, operating model, governance design, financial impact, risk assessment, and a phased transformation roadmap — followed by a detailed account of what was actually built: the architecture, the resilience engineering behind it, and the evidence gathered from testing it end to end, including in a live production deployment.

2. Company Overview

CRED is a trust-first fintech platform built on top of India's premium credit card base. Its original and still-dominant product is credit card bill payment: members pay their card bills through the app and earn reward points for doing so on time, a mechanic that has made the platform synonymous with financially disciplined, high-trust Indian consumers. Around that core, CRED has layered a set of adjacent products — CRED Cash, a short-term credit line drawn against a member's creditworthiness; CRED Mint, a peer-to-peer lending product; CRED Garage, a vehicle-related offering; and CRED Store, a rewards and merchant marketplace.

Strategically, CRED occupies a distinctive position in the Indian fintech landscape: it is not optimising for the mass market on price or reach, but for a premium, single-member experience built around an individual cardholder's trustworthiness. Monetisation today is correspondingly concentrated — card-issuance referral commissions when a member is guided toward a new card, lending interest on CRED Cash and CRED Mint, and merchant-funded rewards partnerships.

The strategic gap this report addresses follows directly from that positioning. CRED's product architecture is built around one person and one login, but Indian financial decisions are rarely made by one person in isolation. A spouse's card, a parent's health cover, a dependent's school fees, and a shared home loan are all part of the same financial picture a household manages jointly — and none of it is visible to CRED today. Closing that gap is the strategic thesis of this report.

3. Industry Dynamics

Four dynamics define the competitive landscape CRED operates in, and together they explain why household-level intelligence is a timely rather than a speculative bet.

UPI as the rail, cards as the margin

The Unified Payments Interface has become the default rail for everyday transaction volume in India, but it carries that volume at a near-zero take rate for the platforms that route it. Credit card issuance, servicing, and the reward economics attached to card spend remain the layer where fintech platforms actually earn margin — which is precisely the layer CRED already owns the deepest relationship in.

Account Aggregator adoption is rising, but under-productised

The Reserve Bank of India's Account Aggregator framework provides licensed, consent-based infrastructure for sharing financial data across institutions. Adoption is growing, but it remains live infrastructure in search of a mainstream consumer product surface — few platforms have built a household-facing experience on top of it. This report treats Account Aggregator integration as a Phase 3 dependency (Section 16) rather than a precondition for the first use case, since Phase 1 does not require it.

Embedded finance is intensifying competition

Jupiter, INDmoney, Fi Money, and a growing set of neobank-adjacent players are converging on the same wealth-and-rewards layer that CRED currently occupies largely alone among premium-positioned players. Differentiation on trust and reward mechanics alone is becoming harder to sustain as these competitors mature.

The household is the under-served unit

No major Indian fintech product treats the family, rather than the individual, as the unit of financial decision-making. This is the specific, defensible gap this report recommends CRED move to close first, ahead of competitors who are still optimising the single-user experience.

4. Current Business Challenges

Stated in business terms, before any technology is proposed, CRED faces four connected challenges.

CHALLENGE	WHAT IS ACTUALLY HAPPENING
Engagement plateaus after bill-pay	Once a bill is paid on time, the member has no structural reason to reopen the app until the next billing cycle. Session frequency is anchored to a monthly event.
Monetization is single-product dependent	Revenue leans heavily on card-issuance commissions. Cross-sell into CRED's other products has been slow to compound into a second meaningful revenue line.
Members transact as a household, CRED sees one person	Spouses, parents, and dependents typically hold separate cards that CRED never links together, so the platform sees a fragment of the household's actual card footprint and spend.
Trust is the brand, but advice is generic	CRED knows a member pays their bills on time. It has no visibility into whether that member is using the right card for a given purchase — the trust relationship is not yet converted into decision-support value.

Each of these four challenges is addressed, at least partially, by the strategy set out in this report. Household-level card routing directly targets the second and third challenges by giving members a reason to return to the app at the moment of a purchase decision, and by requiring the household's full card footprint to be linked for the product to work at all.

5. AI Opportunity Landscape

Seven categories of household financial decision were assessed for where prediction and judgment can materially improve outcomes. Each is summarised below; all seven map onto the seven functional modules of the prototype application (Section 8).

USE CASE	DECISION IT IMPROVES
Card routing	Which card earns the household the most for a given purchase, across every linked member.
Protection gaps	Whether life and health cover is adequate given the household's debt and dependents.
Document intelligence	Extracting key fields from policy and legal documents, and flagging missing or outdated nominees.
Cash-flow advisory	Whether the household can afford a given purchase given its already-committed monthly obligations.
Portfolio concentration	Whether the household's asset allocation is reasonably diversified.
Tax headroom	Section 80C / 80D utilisation and old-vs-new tax regime comparison.
Shared bills	Splitting and tracking obligations across household members transparently.

Card Routing is prioritised first. The reasoning behind that prioritisation is made explicit in the AI Opportunity Matrix below.

5a. Framework — AI Opportunity Matrix

The seven use cases were plotted against two axes: business value to CRED, and feasibility of shipping given current data access and AI risk tolerance. Card Routing is the only use case that scores highly on both axes simultaneously, placing it alone in the Priority Initiatives quadrant.

Why Card Routing wins on both axes:

- *It uses data CRED already has — linked cards, no new consent flow required.*
- *Its pricing logic is fully deterministic, making it the lowest AI-risk-tier use case of all seven.*
- *It produces a useful answer the very first time a household links a second card — no cold-start data problem.*
- *It has a single, directly measurable KPI: rupees of reward value recovered per household per year.*

Protection Gaps and Document Intelligence follow as Longer-Term Bets — high value, but requiring either more sensitive data (insurance and health information) or more mature document-extraction reliability before they are feasible to ship. Portfolio, Tax Headroom, and Cash-flow Advisory sit in the Quick Wins band — useful, but lower business value on their own. Full Bank Integration, which would allow the platform to act on live transaction data rather than typed descriptions, is deliberately placed in the lower-feasibility band: it depends on Account Aggregator infrastructure that is not yet a precondition for launching Card Routing.

6. AI Vision

“Every financial decision a family makes — which card, how much cover, what to file — should have a clear, explainable answer available before the decision is made, not after the statement arrives.”

This vision statement is deliberately constrained by three design principles that recur throughout every module built for this capstone, not just the flagship Card Router.

Household, not individual

Every module reasons across linked household members rather than treating a single cardholder as the unit of analysis. A recommendation that the best card for a purchase belongs to a different family member than the one making the purchase is treated as useful information — a household settlement to be aware of — rather than an edge case to hide.

Explainable by construction

The AI layer explains a decision; deterministic code makes it. This is not a governance policy layered on top of the system after the fact — it is how the system is architected. Every number the application shows is reproducible from code, independent of whether the language model behind the explanation is available at that moment.

Human confirms, human acts

Before anything is priced, the household member sees exactly what the AI understood from their description of a purchase, and can correct it. The model never acts unilaterally, and no module in the prototype ever initiates a payment or transaction.

7. Enterprise AI Strategy

The strategy is sequenced as three bets, ordered from lowest to highest data and regulatory risk, so that each bet’s success is a precondition for approving the next.

BET	WHAT IT PROVES
Bet 1 — Prove judgment at low risk	Card Routing: recommendation only, no autonomous transaction, fully deterministic pricing. This bet is already live.
Bet 2 — Extend to advisory categories	Protection, Tax, and Portfolio: still advisory-only and still human-confirmed, but operating over a broader slice of household data.
Bet 3 — Earn the right to real data	Only once Bets 1 and 2 have demonstrated reliability in production does the roadmap justify the investment in Account Aggregator-linked live data feeds.

7a. Framework — Enterprise AI Canvas

The full strategy is summarised on a single canvas below, following the standard Enterprise AI Canvas structure: customer, value proposition, data, AI capability, integration, use cases, governance, and KPI.

DIMENSION	DEFINITION
Customer	Multi-card households — a primary earner plus linked family members.
Value proposition	Never leave reward value on the table; one household, one answer, regardless of which member's card is best.
Data	Linked card metadata, issuer terms and conditions, and purchase descriptions typed by the user — not scraped or inferred.
AI capability	Natural-language intake (Gemini) combined with a deterministic rules engine of 36 encoded card-terms rules, plus a separate explanation layer.
Integration	An in-app module built on CRED's existing card-linkage flow — no new integration surface required to launch.
Use cases	Card Router (live) → Protection, Tax, and Vault (Phase 2) → Portfolio and Payments (Phase 3).
Governance	Advisory-only risk tier. The household member confirms every parsed purchase before anything is priced.
KPI	Rupees of reward-value leakage recovered per household per year — ₹33,036 as the demonstration-household baseline.

8. AI Use Cases

The prototype implements seven functional modules organised around one household data model. The table below maps each module to the pain point it addresses, the AI capability behind it, and the phase in which it is intended to ship.

MODULE	PAIN POINT	AI CAPABILITY	PHASE
Card Router	Manual reward comparison across cards	NL intake + rules engine + explanation	Phase 1 — live
Protection Score	Unclear whether life/health cover is adequate	Gap analysis against income & dependents	Phase 2
Family Vault	Nominee and estate-planning gaps go unnoticed	Document extraction + legacy-flag detection	Phase 2
Tax Compliance	Unclaimed 80C/80D deduction headroom	Deduction-gap detection + regime comparison	Phase 2
Portfolio	No consolidated household allocation view	Concentration analysis, natural-language Q&A	Phase 3
Cash-flow / Payments	Shared bills tracked manually	Affordability checks, obligation splitting	Phase 3

8a. Use Case #1 — The Live Application: AI Household Card Router

This is the use case built, deployed, and demonstrated for this capstone. It deserves a detailed account, because the way it is built is as much a part of the strategic argument as the business case for building it.

The problem in concrete terms

A household holding four or five credit cards leaves real reward value on the table, because the terms that decide how much a purchase earns are buried in fine print nobody reads at the point of payment. Three examples, drawn directly from the card terms encoded in the prototype, illustrate why this is a genuinely hard problem rather than a simple lookup:

- The same flight can earn 16.5% booked through a card’s rewards portal, and only 3.3% booked directly with the airline — a more than fourfold difference driven entirely by booking channel, a detail no dropdown menu naturally captures.
- A headline “5% cashback on all online spend” offer is frequently capped at a fixed monthly rupee amount. A single large purchase can silently blow through that cap and earn a fraction of the advertised rate for the remainder of the transaction.
- On international spend, a 3.5% foreign-transaction markup can exceed the reward earned outright — meaning the card advertised as the household’s best “rewards card” can actively lose money on a specific purchase.

Existing personal-finance and card-comparison tools optimise for one person holding one card. None of them route across the members of a household, which is precisely the layer CRED is positioned to own because it already has the card-linkage relationship with each member individually.

How the application is built

The architecture is deliberately split into three parts that never share responsibility for the same decision, and this separation is the core engineering argument of the whole capstone:

- **Natural-language intake.** The household member describes a purchase in plain English — for example, “Booking an ₹80,000 flight to Dubai through SmartBuy.” A language model (Google Gemini) extracts the amount, spend category, merchant, and, critically, the booking channel. Channel is language-only information: “through SmartBuy” versus “direct with the airline” cannot be expressed by a structured form field, yet it is exactly the detail that decides the outcome.
- **Human confirmation.** Before anything is priced, the parsed purchase — amount, category, merchant, channel — is displayed back to the member for correction. Nothing is priced until the member confirms or edits what the model understood.
- **Deterministic pricing.** A rules engine, encoding 36 individually labelled terms-and-conditions rules across five card products, prices every card in the household against the confirmed purchase. Every card’s value is computed by ordinary application code — the language model is never asked to perform, and never performs, the arithmetic that decides which card wins.
- **AI explanation.** Once the winner is determined by code, a second, narrower model call turns the rules engine’s decision trace into plain-language reasoning, citing the specific clause that decided the outcome.

Verified behaviour — evidence, not assertion

The following results were produced by the deployed application, not simulated for this report. They are reproducible by re-running the same inputs against the live URL.

TEST	RESULT
₹80,000 flight, booked “through SmartBuy”	HDFC Infinia Metal wins at ₹13,200 (16.5% effective rate) — the portal-accelerator rule fires.
Same ₹80,000 flight, booked “direct with the airline”	Axis Atlas wins at ₹4,000 (5% effective rate) — the portal rule never applies, a ₹9,200 swing from one phrase.
₹2,00,000 online purchase	HDFC Infinia (3.3% base rate, ₹6,600) beats SBI Cashback, whose advertised 5% rate is capped at ₹5,000/month and would otherwise have looked like the obvious winner.
₹45,000 rent payment	Every card in the household excludes rent from rewards. The application returns an honest “no card earns rewards here” verdict instead of inventing a winner.
₹1,20,000 spend in Dubai	Axis Atlas returns a negative net value (–₹1,800): the 3.5% forex markup exceeds the 2% base reward rate on that card.

Every one of these results includes a full decision trace in the application UI: every rule the engine evaluated, whether it fired, and the exact fine-print clause responsible — so the recommendation is auditable rather than a black box.

Resilience engineering

A live demonstration in front of a Board panel does not get a second attempt, so the application was built with two tiers of failure handling rather than an optimistic assumption that the AI provider will always respond.

- **Automatic model and key failover.** Both the API key and the model name accept comma-separated lists. Every call walks that list and uses the first candidate that answers, so a single rate-limited model or exhausted key never surfaces as a user-facing error — the system simply moves to the next candidate. This was proven directly during development: a deliberately exhausted model was placed first in the failover chain, and the application transparently fell through to a working model and returned a correct, high-confidence result.
- **Deterministic fallback.** If every configured model is unreachable, purchase intake falls back to a keyword-based parser and the explanation falls back to rules-only prose generated directly from the decision trace. Both fallback states are visibly labelled in the interface, so a degraded response is never mistaken for the model's full-quality output — and, crucially, the recommendation itself is unaffected, because it never depended on the model in the first place.

This resilience design directly answers a question a technically literate Board member (playing the Chief Digital Officer role) is likely to ask: what happens when the AI provider fails during the live demo? The honest answer is that it already has, during testing, and the application recovered without any visible degradation in the recommendation quality.

9. AI Operating Model

Approving Card Router changes how four groups within CRED work, not just what the product does.

OWNER	RESPONSIBILITY
Product	Owns the use-case backlog against the AI Opportunity Matrix, not an open feature-request queue.
AI / ML	Owns model selection, failover configuration, and ongoing explanation-quality review.
Risk & Compliance	Owns the risk-tier classification and sign-off required before any new use case ships.
Household member	Retains the final action in every interaction — confirms every AI reading before it is priced or acted on.

The decision rights embedded in this model are deliberately narrow: the AI model never has the authority to move money or finalise a recommendation without the member’s confirmation step. This is not a policy statement layered on top of the product after launch — it is enforced by the architecture described in Section 8a.

9a. Framework — AI Operating Model Blueprint

Five layers, each with a single accountable owner, describe how the system is built and run end to end.

LAYER	OWNER	RESPONSIBILITY
Strategy	Product	Which use case ships next, sequenced by the Opportunity Matrix.
Data	Data Engineering	Card linkage, issuer terms-and-conditions ingestion, and consent design (Account Aggregator framework, future).
Model	AI / ML	Intake and explanation models, and the failover chain across models and API keys.
Application	Engineering	The deterministic rules engine, the decision trace, and the UI confirmation step.
Governance	Risk & Compliance	Risk tiering, human-in-the-loop enforcement, and the audit trail.

10. Data Strategy

The prototype and a production deployment differ sharply in their data footprint, and this report is explicit about the gap rather than glossing over it.

Today — prototype

- Two synthetic households, compiled directly into the application — there is no external data source or persistent database in the deployed instance.
- 36 card terms hand-encoded from published, publicly available fee schedules.
- Purchase details are typed by the user; there is no live transaction feed.

- No personally identifiable information is collected by the deployed application.

Production — required

- Consented transaction and card data accessed via the RBI-licensed Account Aggregator framework.
- An issuer terms-and-conditions ingestion pipeline with version tracking, since card terms change without notice.
- Card-on-file tokenisation, so spend can be routed against real transactions rather than typed descriptions.
- A retention and consent design compliant with the Digital Personal Data Protection Act, 2023, reflecting that this data now spans an entire household rather than one individual.

This gap is treated as a Phase 3 dependency in the transformation roadmap (Section 16), not a blocker to approving Phase 1 — Card Router does not require any of the production-grade data infrastructure listed above to launch.

11. Technology Architecture

Prediction and decision are deliberately implemented as separate systems, connected by a human confirmation step. This section describes the pipeline in the order a purchase actually moves through it.

STAGE	WHAT HAPPENS
1. Plain English	The household member types the purchase description exactly as they would say it out loud.
2. AI intake	Gemini extracts the amount, spend category, merchant, and booking channel from the sentence.
3. Human confirms	The parsed reading is shown for correction. Nothing is priced until the member confirms it.
4. Rules engine	36 encoded rules price every card in the household deterministically, in ordinary application code.
5. AI explains	A second model call narrates the decision the rules engine already made, citing the exact clause responsible.

Cost to run

Each recommendation costs under ₹1 to generate, using two short model calls at current flash-tier pricing. The pricing engine itself — the component that actually decides the rupee value of every card — is deterministic code and costs nothing per call, so the marginal cost of the product scales with how often people use it, not with how many cards, rules, or households are added to the system.

Resilience, tested in production

- A model-and-key failover chain, so a rate-limited model moves to the next candidate automatically rather than surfacing an error to the user.
- A per-attempt timeout, so a hung model call fails fast rather than stalling silently — a real risk observed during development, where one candidate model hung indefinitely rather than returning an error.
- A deterministic fallback path, so that if every configured model is unreachable, the application still returns a correct, honestly labelled, rules-only recommendation.

12. AI Governance

Decision rights in this system are narrow by design, not by policy overlay. The distinction matters: a governance document that says a model should not act unilaterally is a promise; an architecture in which the model has no code path to act unilaterally is a guarantee.

WHAT THE MODEL MAY DO	WHAT THE MODEL MAY NEVER DO
Read a purchase description	Choose the winning card unilaterally
Price cards using disclosed, deterministic logic	Move money or initiate a transaction
Explain a decision that has already been made	Skip the human confirmation step

The application’s risk tier is Advisory / Recommendation — the lowest tier on the Risk Assessment Matrix in Section 15a — because no code path exists for the model to move funds or finalise a decision without a human step in between.

12a. Framework — AI Governance Canvas

DIMENSION	DESIGN CHOICE
Decision authority	Recommendation only. The final action is always taken by the household member.
Risk tier	Advisory / Low — no funds movement, no regulated credit decision.
Human-in-the-loop point	The parsed purchase (amount, category, channel) is shown for correction before pricing.
Explainability requirement	Every recommendation ships with a full, rule-by-rule decision trace.
Escalation path	Ambiguous or all-excluded cases surface an honest “no good option” verdict, never a guess.
Model failure handling	Falls back to deterministic, rules-only pricing — the recommendation is never blocked by a model outage.

13. Business Value Framework

Value is created at four distinct points, only one of which is the headline recovered rupee.

VALUE DRIVER	DESCRIPTION
Direct member value	₹33,036 per year recovered per multi-card household — the headline, member-visible number.
Engagement	A reason to reopen the app before the next billing cycle, directly addressing the engagement plateau described in Section 4.
Data asset	Every linked card and confirmed purchase deepens CRED’s view of the household’s real spending pattern.

VALUE DRIVER	DESCRIPTION
Cross-sell surface	A household with visible, quantified reward leakage is a qualified lead for its next card — the same monetisation mechanism CRED already runs today.

CRED does not need to capture the recovered rupee directly to benefit from this product. The value-capture mechanism is engagement and card-issuance referral — both existing revenue lines — applied to a stickier, more frequent product surface than bill payment alone provides.

14. Financial Impact

The figures in this section are illustrative sizing built bottom-up from the live application's demonstration-household economics, for the purposes of this class exercise. They are explicitly not sourced from, and should not be read as, any disclosure of CRED's actual financial results, in line with the academic integrity requirement to mark estimates clearly.

Illustrative value pool

REACH (ILLUSTRATIVE)	GROSS ANNUAL VALUE POOL (₹ CR/YR)
100,000 households	₹3.3 Cr
500,000 households	₹16.5 Cr
1,000,000 households	₹33.0 Cr
2,000,000 households	₹66.0 Cr

Calculated as ₹33,036 per household per year × illustrative reach. This is a scaling exercise on the demonstration household's own economics, not a projection of CRED's actual user base.

Cost structure

- Under ₹1 per recommendation served — two short model calls at current flash-tier pricing.
- ₹0 incremental cost per additional card or rule added to the engine, since pricing logic is ordinary code, not model inference.

Take-rate logic

CRED does not need to capture the recovered rupee directly. The mechanism by which this initiative converts to revenue is engagement and card-issuance referral commission — both existing, proven revenue lines — applied to a product surface a household will use far more often than once-a-month bill payment.

15. Risks

Four risks are named explicitly, each with a mitigation that is already built into the system rather than planned for a future release.

RISK	MITIGATION
Hallucinated pricing	Impossible by construction — the language model never computes a rupee figure; deterministic code does.
Provider rate limits during a live demo	Automatic failover across models and API keys, verified in production by deliberately exhausting a model and observing correct recovery.
Stale or incorrect card terms	Terms are sourced from published schedules today; production requires a version-tracked ingestion pipeline, flagged explicitly as a Phase 2 requirement.

RISK	MITIGATION
Bias toward well-documented cards	Cards without encoded terms default to a conservative 1% base rate and are flagged at low confidence, rather than silently guessed at a higher rate.

15a. Framework — Risk Assessment Matrix

Plotting these risks, plus two additional operational risks, against likelihood and impact produces the following prioritisation.

RISK	LIKELIHOOD	IMPACT	STATUS
Hallucinated pricing	Low	High (if it occurred)	Eliminated by design, not merely mitigated
Provider rate limits mid-demo	Medium-High	Medium	Multi-model failover live and tested
Stale card terms	Medium	Medium	Flagged as a Phase 2 pipeline requirement
Provider outage	Low-Medium	Medium	Covered by the same failover chain
Ambiguous purchase intake	Medium-High	Low	Surfaced to the human for confirmation, never guessed silently

16. Transformation Roadmap

The roadmap is phased in three stages, and the first stage is not a future commitment — it is already shipped.

PHASE	TIMEFRAME	FOCUS	KEY MILESTONES
Phase 1	0–6 months	Card Router	Deploy to CRED’s member base · Instrument the rupee-recovered KPI · Add a real card-linkage data feed. Status: live today.
Phase 2	6–12 months	Advisory suite	Ship Protection Score, Family Vault, and Tax Compliance · Reuse the same advisory-only risk tier and human-confirm pattern.
Phase 3	12–24 months	Real data, full household	Account Aggregator integration · Portfolio and Payments modules · Household-level consent design.

17. Future Vision

Three to five years out, the ambition is not an app with modules bolted on, but a household financial operating system.

Every decision, explained

Card, cover, tax, and investment choices all carry a visible, auditable rationale — the decision-trace pattern proven in Card Router, generalised across every module.

Real data, real consent

Account Aggregator-linked live feeds replace typed purchase descriptions and synthetic demonstration households, once Phases 1 and 2 have proven the product and governance model can be trusted with that data.

CRED as the household layer

Not a bill-pay app with modules attached, but the default surface a family opens whenever it needs to decide anything financial — the natural endpoint of the strategy set out in this report.

18. Final Recommendations

Three specific approvals are requested from the Board.

#	RECOMMENDATION	WHY NOW
1	Approve Card Router for member rollout	It is built, deployed, and tested — approval starts a rollout, not a build.
2	Fund the Phase 2 advisory modules	They reuse the same architecture and risk tier as Card Router; Protection, Vault, and Tax are additive scope, not new risk.
3	Commission the Account Aggregator integration	It is the single dependency standing between this prototype and access to real household data.

A strategy without a working prototype is a proposal. This is not a proposal.

19. Application Brief

This section reproduces, for completeness within this report, the required one-page Application Brief describing the submitted use case, AI capability, data inputs, key output, business KPI, and known limitations.

Use case

This application determines which credit card in a household earns the most on a given purchase — across every family member — so the family captures reward value it currently loses to fine print it never reads at the point of payment.

AI capability used

Google Gemini performs two distinct roles: natural-language intake, which drives the application's core output by extracting amount, category, merchant, and booking channel from a plain-English description; and an explanation layer, which narrates a pricing decision made entirely by deterministic code. The model never computes a rupee value and never selects the winning card.

Data inputs

Household members, incomes, and linked cards (synthetic in the prototype); 36 hand-encoded card terms-and-conditions rules; purchase details typed in natural language by the user; and a representative monthly spend basket per household, used to compute the rewards-leakage KPI.

Key output

The winning card and its owner, with net rupee value after exclusions, caps, and forex costs; runner-up cards and the exact value forgone by choosing them; a full rule-by-rule decision trace; a confidence rating; fee-waiver milestone impact; and a cross-member settlement note when the best card belongs to a different household member than the one making the purchase.

Business KPI it addresses

Rewards leakage — reward value forgone through suboptimal card selection. For the demonstration household, the application computes ₹33,036 per year of leakage, derived live from the household's own spend basket rather than asserted as a fixed figure.

Known limitations

- Card terms are encoded from published schedules and simplified for this prototype; issuers revise them without notice, and there is no ingestion pipeline yet to track revisions.
- Reward-point valuations are estimates; transfer-partner and promotional bonuses are excluded.
- Monthly caps assume no prior spend in the current billing cycle, since the prototype has no visibility into spend already made.
- Households are synthetic; no live bank or card data is connected.
- If the language model is unavailable, intake and explanation degrade to deterministic fallbacks, both clearly labelled in the interface; the recommendation itself is unaffected.
- This is a spending suggestion for a human to act on — not financial advice, and no payment is ever initiated by the application.

20. Academic Integrity & Disclosures

In accordance with the capstone's academic integrity requirements, the following disclosures apply to this report and the accompanying application and strategy deck.

- AI-generated code was used throughout the application's development, consistent with the capstone's vibe-coding brief. The AI coding assistant used was Claude (Anthropic).
- Company, market-sizing, and financial figures in Sections 14 and elsewhere are explicitly illustrative estimates built bottom-up from the demonstration application's own data, and are not sourced from or intended to represent CRED's actual disclosed financials.
- Industry and competitive context in Section 3 reflects general, publicly known dynamics in the Indian fintech sector as of the time of writing and should be treated as directional rather than precisely sourced market research.
- All household, member, card, and transaction data used in the application and in this report is synthetic and was constructed for illustrative purposes only.

21. Appendix — Live Deployment & Repository

Live application: <https://dtai-fintech-cred.vercel.app>

GitHub repository: <https://github.com/prathameshkumbhar1206-alt/DTAI-FINTECH-CRED>

The repository includes the full application source, the README documenting local setup and deployment steps, the 25-slide Board strategy deck (PDF and PPTX), and this report. The application is deployed on Vercel and requires no database or persistent infrastructure — all demonstration households are compiled directly into the application bundle, which is itself a deliberate architectural choice explained in Section 10.

End of report.